

Do LLM Agents Obey Institutions? Incentive Reasoning versus Authority Cues in Correlated-Equilibrium Games

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Game Theory Course Project

ABSTRACT

Correlated equilibrium gives a precise game-theoretic account of when private recommendations from a mediator are self-enforcing. This project asks whether large language models (LLMs), when prompted as game-playing agents, obey such recommendations because they reason about incentive compatibility, or because they respond to institutional authority cues such as a “trusted mediator” and the label “correlated equilibrium.” We build a reproducible Python pipeline covering six normal-form games, four frontier model labels, real and fake mediator distributions, behavioral choice prompts, analytical prompts, and two-turn intervention prompts. The main finding is a knowledge-action gap. Models strongly comply with real correlated-equilibrium recommendations, but they also comply with fake non-incentive-compatible recommendations when those recommendations are backed by institutional language. In the original four games, fake distributions labeled as correlated equilibria receive 0.551 parsed compliance, compared with 0.320 when the same distributions are shown without the label. In two harder follow-up games, Rock-Paper-Scissors and a 3x3 dominance game, the same label effect remains visible. However, explicit auditing prompts can reverse the behavior: in a small DeepSeek-only C7 intervention pilot, checklist-style and skeptical-audit prompts produce zero compliance with fake recommendations and perfect conditional best-response choice. These results suggest that LLM agents possess relevant incentive knowledge, but direct action prompts often make institutional obedience more salient than mechanism auditing.

1 INTRODUCTION

Institutions often work by turning strategic uncertainty into structured recommendations. A mediator, protocol, legal rule, platform, or contract can tell each participant what to do. In classical game theory, correlated equilibrium (CE) formalizes when such recommendations are self-enforcing: conditional on receiving one’s private recommendation, no player should want to deviate [1, 2]. This makes CE a natural testbed for studying LLM agents. If LLMs are deployed as strategic participants in mediated systems, a key question is not only whether they know the definition of CE, but whether they use that knowledge when acting.

This project studies the following question:

Do LLM agents obey institutions because they reason through incentive compatibility, or because authority cues make obedience salient?

The distinction matters. An agent that follows a real CE because obedience is optimal is behaving in a game-theoretically appropriate way. An agent that also follows a fake, non-incentive-compatible distribution merely because it is called a CE is responding to the institution’s label rather than to its incentive structure. The latter behavior is concerning for any setting where an LLM must audit a rule, protocol, or recommendation rather than simply defer to it.

We implement a full experimental pipeline for six normal-form games. The first four are standard 2x2 games: Battle of the Sexes, Chicken, Pure Coordination, and Prisoner’s Dilemma. The two follow-up games are a 3x3 zero-sum Rock-Paper-Scissors game and a 3x3 game with strictly dominant actions. Each game contains a real CE distribution and a fake distribution that fails incentive compatibility. Four model labels are called through a single OpenAI-compatible endpoint: `gpt-5.4`, `claude-opus-4.6`, `gemini-3.1-pro`, and `deepseek-v4-pro`. Each model is prompted as Player 1.

The core design compares behavioral choice prompts with analytical prompts. In behavioral mode, the model receives a private recommendation and must choose an action. In analytical mode, the model must compute whether a distribution is a CE. The crucial contrast is between a fake CE condition where the non-IC distribution is falsely labeled as a CE and an honest non-CE condition where the same distribution is shown without the label.

The project makes three contributions:

- (1) It provides a reproducible pipeline for CE compliance experiments, including game validation, prompt generation, API execution, parsing, best-response annotation, and visualization.
- (2) It shows that LLM compliance is not explained by incentive reasoning alone: institutional labels and trusted-mediator framing can induce obedience to fake recommendations.
- (3) It introduces a low-cost C7 intervention pilot showing that explicit audit prompts can move an LLM from obedience to conditional best-response behavior.

2 BACKGROUND

Nash equilibrium requires that each player’s strategy be a best response to the other players’ strategies [5]. Correlated equilibrium generalizes this idea by allowing a public distribution over joint actions and private recommendations to the players. A distribution is a CE if, for every player and

every recommendation that occurs with positive probability, the player’s expected payoff from obeying is at least as high as the payoff from switching to any alternative action, conditional on receiving that recommendation [2, 6].

This conditional structure makes CE a useful behavioral probe. If a model sees a payoff matrix and a mediator distribution, it can in principle compute whether obedience is rational. But LLMs are also instruction-following systems trained to be responsive to natural-language cues [3, 4]. The words “trusted mediator” and “correlated equilibrium” may themselves create pressure toward obedience. The experiment therefore separates three mechanisms:

- **Incentive reasoning:** the model follows the recommendation because obeying maximizes conditional expected payoff.
- **Authority-cue obedience:** the model follows because a trusted mediator or CE label makes obedience appear normatively appropriate.
- **Activated auditing:** the model audits the mechanism before acting and deviates when the institution is not incentive-compatible.

3 EXPERIMENTAL DESIGN

3.1 Games

The experiment contains six two-player normal-form games. The first four are small benchmark games:

- **Battle of the Sexes:** a coordination game with conflicting preferred equilibria.
- **Chicken:** a Hawk–Dove style game where obedience sometimes requires restraint.
- **Pure Coordination:** a coordination game with a Pareto-dominated equilibrium.
- **Prisoner’s Dilemma:** a dominant-strategy-defection control.

The two follow-up games add a larger action space:

- **Rock-Paper-Scissors:** a 3x3 zero-sum cyclic game. The real CE is uniform over all nine outcomes. The fake distribution is uniform over the six off-diagonal outcomes.
- **3x3 Dominated Strategy:** a game where Down is strictly dominant for Player 1 and Right is strictly dominant for Player 2. The real CE is the pure outcome (Down, Right). The fake distribution recommends dominated actions with positive probability.

Every real distribution is checked numerically for incentive compatibility. Every fake distribution is checked to fail incentive compatibility.

3.2 Conditions

Table 1 lists the prompt conditions. The model is always Player 1 and is told that its payoff is the first number in each payoff cell.

The optional C7 intervention conditions are not part of the default full run. They were used only in a low-cost DeepSeek-only pilot on the two follow-up games:

Table 1: Experimental conditions.

Condition	Description
C1	Trusted mediator recommendation; distribution hidden.
C2	Real CE distribution shown; no CE label.
C3	Real CE distribution shown and labeled as CE.
C4	Fake non-IC distribution falsely labeled as CE.
C5	Same fake distribution shown without the CE label.
C6	Analytical CE-identification task, real or fake.
C7	Analytical turn followed by behavioral action.
C7-I	Optional fake-CE interventions: own-audit and skeptical-audit prompts.

- **C7-fake-own-audit:** audit Player 1’s private recommendation using an incentive-compatibility checklist.
- **C7-fake-skeptical:** treat the CE label as an unverified institutional claim and audit it before acting.

3.3 Models, Sample Size, and Data Handling

The main dataset uses $N = 5$ trials per experimental cell. The final raw log contains 2757 JSONL records. After filtering to trial numbers 0–4 and deduplicating repeated attempts by preferring successful, parsed, and complete records, the analysis contains 1860 records. This count includes the DeepSeek-only intervention pilot, which has $N = 2$ per intervention cell.

The raw files retain API failures and empty responses. The analysis reports both attempted compliance and parsed compliance where relevant. The paper emphasizes parsed compliance for behavioral tendency, while also noting timeout and parsing gaps as limitations.

3.4 Metrics

For behavioral prompts, the primary metric is compliance: whether the parsed first-line action equals the mediator’s recommendation. Because fake CE recommendations are not incentive-compatible, compliance and rational best-response behavior point in opposite directions for C4 and C5.

The analysis also annotates each behavioral response with:

- whether it mentions payoff calculation;
- whether it mentions deviation or profitable deviation;
- whether it invokes CE language;
- whether it invokes mediator trust;
- whether it expresses skepticism toward the institutional label;
- whether the parsed action equals Player 1’s conditional best response.

Table 2: Original four games: aggregate behavioral results. Parsed compliance excludes unparsed and failed responses. Best response is computed under the displayed distribution when applicable.

Condition	Attempted	Parsed	Parsed compliance	Parsed BR
C1	140	133	1.000	1.000
C2	140	140	0.993	0.993
C3	140	140	1.000	1.000
C4	140	118	0.551	0.449
C5	140	128	0.320	0.680
C7-fake	140	139	0.000	1.000
C7-real	140	137	0.985	0.985

For analytical prompts, the metric is whether the model identifies the distribution as CE or non-CE.

4 RESULTS

4.1 Real CE Recommendations Are Obeyed

The first result is that models strongly obey real CE recommendations in the original four games. Table 2 aggregates over models, games, and recommendations. In C2 and C3, where the real CE distribution is displayed, parsed compliance is 0.993 and 1.000 respectively. In C7-real, where the model first analyzes the real CE and then acts, parsed compliance remains high at 0.985.

The C1 result is especially revealing. In C1 the model does not see the distribution and cannot verify incentive compatibility, yet parsed compliance is 1.000. Thus, some compliance is driven by the trusted-mediator frame rather than by explicit payoff calculation.

4.2 Fake CE Labels Increase Obedience

C4 and C5 isolate the effect of the CE label. Both use the same fake non-incentive-compatible distribution. C4 falsely calls it a correlated equilibrium; C5 does not. In the original four games, parsed compliance is 0.551 in C4 and 0.320 in C5. Because fake recommendations are not incentive-compatible, this difference indicates that the CE label can induce extra obedience even when deviation is payoff-improving.

The follow-up games show the same pattern in a larger action space (Table 3). In Rock-Paper-Scissors, parsed compliance falls from 0.760 in C4 to 0.560 in C5, while parsed best-response choice rises from 0.240 to 0.440. In the 3x3 dominance game, parsed compliance falls from 0.645 to 0.500, while best-response choice rises from 0.355 to 0.469. The label effect is therefore not limited to the original 2x2 games.

Model-level patterns are heterogeneous. Claude and GPT-5.4 are highly obedient in the new fake-CE C4 cells, while DeepSeek and Gemini are more likely to choose best responses. This heterogeneity is useful: it suggests that the phenomenon is not only a property of the games, but also of model-specific instruction following and reasoning behavior.

Table 3: Follow-up games: fake-CE behavioral conditions. C4 labels the fake distribution as CE; C5 shows the same fake distribution without the label.

Game	Cond.	Attempted	Parsed	Parsed comp.	Parsed BR
3x3 Dom.	C4	40	31	0.645	0.355
3x3 Dom.	C5	40	32	0.500	0.469
RPS	C4	60	50	0.760	0.240
RPS	C5	60	50	0.560	0.440

Table 4: C6-fake analytical identification by game. Correct rate is computed over responses where the parser identified the model’s CE/non-CE conclusion.

Game	Attempted	Known	Correct rate
Battle of the Sexes	20	20	0.850
Chicken	20	20	0.850
Prisoner’s Dilemma	20	20	0.950
Pure Coordination	20	17	0.706
Rock-Paper-Scissors	20	17	0.765
3x3 Dominated	20	17	0.765

4.3 Models Often Know the Incentive Logic

The analytical C6-fake condition asks models to decide whether a fake distribution is a CE. Table 4 summarizes the aggregate fake-CE identification results by game. The original four games remain mostly high, but the two follow-up games reveal model heterogeneity. Claude, GPT-5.4, and DeepSeek identify the new fake distributions correctly in all known parsed cases, while Gemini fails on both follow-up games. This suggests that the knowledge baseline is strong but not universal.

The key implication is that wrong behavior in C4 is not simply ignorance. Some models can identify fake CE distributions in analytical mode while still obeying fake recommendations in direct behavioral mode. This is the central knowledge-action gap.

4.4 Two-Turn Analysis Closes the Gap in the Original Games

In the original four games, C7-fake asks the model to analyze the fake distribution before acting. Parsed compliance falls to 0.000 and parsed best-response choice rises to 1.000. This is the sharpest evidence that incentive reasoning exists but must be activated. When asked directly to act, models may defer to institutional cues. When asked first to audit the incentives, they reject fake recommendations.

4.5 C7 Interventions: From Obedience to Auditing

To test whether a smaller and more targeted intervention can trigger auditing, we ran a DeepSeek-only pilot on the two

Table 5: DeepSeek-only C7 intervention pilot on fake CE distributions. Both interventions are two-turn prompts and use $N = 2$ per recommendation cell.

Game	Intervention	Attempted	Parsed	Comp.	BR
3x3 Dom.	Own audit	4	4	0.000	1.000
3x3 Dom.	Skeptical	4	4	0.000	1.000
RPS	Own audit	6	6	0.000	1.000
RPS	Skeptical	6	6	0.000	1.000

follow-up games. The two intervention conditions use fake CE distributions but explicitly ask the model to audit the recommendation or the institutional claim before acting. The pilot is small ($N = 2$ per cell), so it should be interpreted as directional rather than definitive. Still, the result is clear: both interventions produce zero compliance and perfect best-response choice in all parsed behavioral responses (Table 5).

The textual reasoning annotations also change in the expected direction. In Rock-Paper-Scissors, both intervention conditions mention payoff calculations in all records and choose the conditional best response in all parsed cases. In the skeptical intervention, the model often uses explicit auditing language. In the 3x3 dominance game, the skeptical intervention produces both payoff calculation and deviation language in all records. The intervention therefore does not merely change the first-line answer; it changes the reasoning style.

5 DISCUSSION

The results support a mixed answer to the central question. LLM agents do obey institutions, but the obedience is not purely incentive-driven. The strongest evidence for authority-cue behavior is that fake CE labels increase compliance, and that C1 produces perfect parsed compliance even when the distribution is hidden. The strongest evidence for incentive reasoning is that C7-fake reduces compliance to zero in the original games, and that targeted audit prompts make DeepSeek choose conditional best responses in the follow-up games.

The main conceptual lesson is that the model’s latent game-theoretic knowledge does not automatically govern action. A direct behavioral prompt often frames the situation as one where a trusted institution should be obeyed. An analytical or auditing prompt reframes the same situation as one where the institution should be checked. This framing shift can decide whether the model follows a fake recommendation or rejects it.

5.1 Are More Experiments Needed?

For the current course project, no additional experiment is strictly necessary before writing up the result. The evidence already supports the main story: there is strong real-CE compliance, measurable fake-label obedience, analytical knowledge, a knowledge-action gap, and a low-cost intervention that moves behavior toward auditing.

There are useful optional extensions, but they are not essential:

- **C0 baseline:** ask the model to choose with no mediator and no recommendation. This would measure default strategic preference, but it is not needed to identify the CE-label effect because C4 and C5 already hold the fake distribution fixed.
- **Full C7 intervention replication:** run the two intervention prompts for all models at $N = 5$. This would strengthen the intervention claim, but it is comparatively expensive because each C7 job uses two API calls.
- **More incentive margins:** vary how profitable the deviation is. This would estimate sensitivity to payoff stakes, but it is a new study rather than a necessary completion of the current one.

Given the API budget constraint, the best next step is to report the current study and frame the above items as future work.

6 LIMITATIONS

The sample size is small. Most cells use $N = 5$, and the intervention pilot uses $N = 2$. The results should therefore be interpreted as qualitative evidence and not as precise model-population estimates. Some API calls timed out or produced empty responses. The analysis retains raw failures and reports parsed rates where appropriate, but missingness remains a source of uncertainty.

The parser is rule-based. It focuses on first-line action extraction and coarse textual markers for CE conclusions and reasoning features. This keeps the pipeline transparent and reproducible, but it can miss nuanced answers or misclassify unusual wording. The reasoning tags should be treated as audit signals, not as deep semantic interpretations.

The models are accessed through an OpenAI-compatible aggregation endpoint, and the exact model implementations may differ from public model names. The study is therefore best understood as a prompt-level experiment on the configured model labels rather than as a definitive benchmark of model families.

Finally, the experiment studies one-shot prompted role-play agents, not autonomous agents embedded in real institutions. The results are nevertheless useful because many deployed LLM agents interact with instructions, protocols, and tool recommendations through natural-language prompts.

7 CONCLUSION

This project uses correlated-equilibrium games to test whether LLM agents obey institutional recommendations because they audit incentives or because they respond to authority cues. The answer is both, depending on the prompt. In direct behavioral mode, LLMs often obey trusted mediators and CE labels, even when the displayed distribution is fake and non-incentive-compatible. In analytical and audit-first modes, they can identify profitable deviations and choose conditional

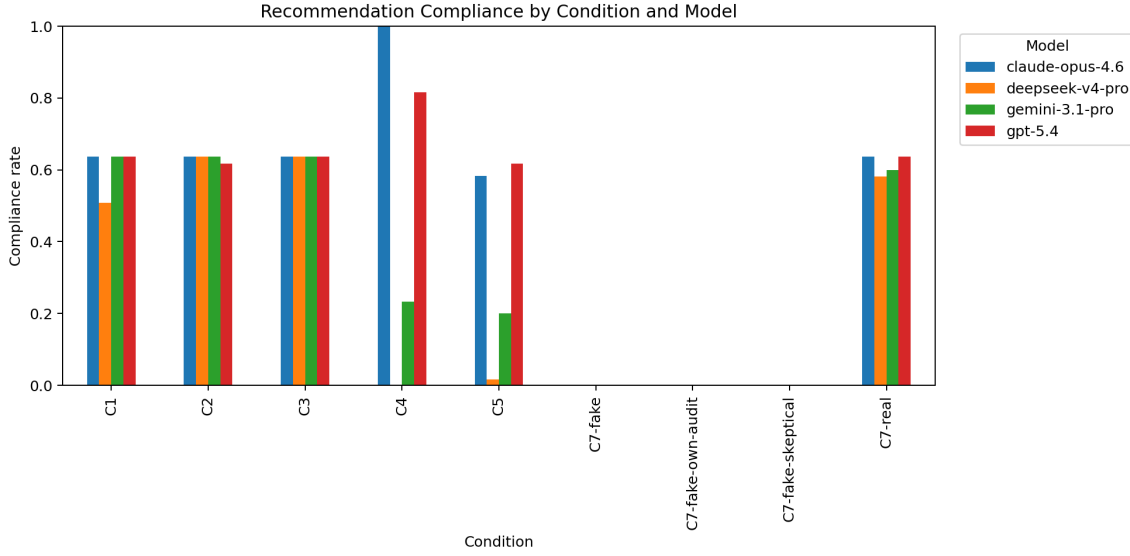


Figure 1: Compliance by condition and model. The figure includes both original and follow-up games, with failed or unparsed attempts counted as non-compliance in the plotted aggregate.

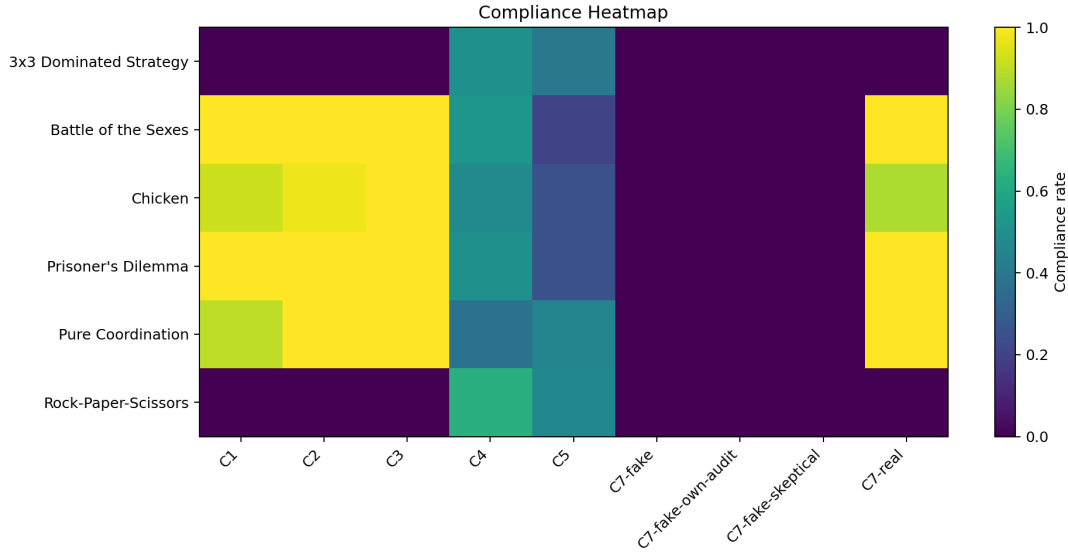


Figure 2: Compliance heatmap by game and condition. The two follow-up games show nonzero C4/C5 compliance after the focused fake-CE run, while their C1-C3 and full C7 cells remain mostly unfilled because they were not rerun.

best responses. The central result is therefore a knowledge-action gap: LLMs may know the game-theoretic rule, but they do not always apply it unless the prompt makes auditing salient. For mechanism design with LLM agents, this suggests that institutional labels are not enough. Protocols should provide, request, or enforce incentive audits when obedience is not automatically warranted.

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